

Martin Daigne

Robotics Master's Student — Data Science Minor

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Education

MSc	École Polytechnique Fédérale de Lausanne (EPFL) , Robotics - Data Science minor	Switzerland
	Core courses: Machine Learning, Deep Learning, Reinforcement Learning, Convex Optimization, Model Predictive Control, Learning and Adaptive Control for Robots, Mobile Robotics, Robotics for Manipulation, Applied Data Analysis, Modern Natural Language Processing...	2024 – 2027
	Delft University of Technology (TU Delft) , Exchange - Mechanical/Aerospace Engineering	Netherlands
		2023 – 2024
BSc	École Polytechnique Fédérale de Lausanne (EPFL) , Mechanical Engineering	Switzerland
	- Grade: 5.57/6 - Core courses: Analysis, General Physics (Mechanics, Thermodynamics, Electromagnetics), Numerical Analysis, Dynamical Systems, Continuum Mechanics, Structural Mechanics, Nonlinear Mechanics, Fluid Mechanics, Experimental Fluids, Heat and Mass Transfer, Control Systems, MEMS, Intelligent Vehicles...	2021 – 2024

Experience

EPFL, Laboratory of Intelligent Systems (LIS) , Semester Project	Switzerland
Design and implementation of diffusion policies (DDPM, DDIM, Score Matching, Flow Matching) to generate control trajectories for a morphing drone previously controlled with an MPPI.	Feb 2026 – June 2026 5 months
EPFL, Computational Robot Design and Fabrication Lab (CREATE) , Semester Project	Switzerland
Data-Enabled Model Predictive Control for Compliant Manipulators: I built a simulation in Python and an adaptive MPC controller for a soft, tendon-driven arm using CasADi and acados for real-time deployment.	Sept 2025 – Jan 2026 5 months
EPFL, Unsteady Flow Diagnostics Laboratory (UNFoLD) , Summer in the Lab Internship	Switzerland
Design, building, and analysis of a random gust generator to improve realism in wind turbine experiments.	June 2024 – Aug 2024 3 months
Geneva University (UNIGE), Athena Program , Student Researcher	Switzerland
1st semester: Set Theory and Logic in Mathematics alongside high school. 2nd semester: Research on Kuratowski's Theorem in Set Theory.	Sept 2020 – June 2021 10 months
Aether Swiss Kite , Control Team Leader	Switzerland
Lead the data acquisition, system identification, and control of the kite to achieve optimized automated control.	2025 – 2026 1 year
Aether Swiss Kite , Electrical and Control Engineer	Switzerland
1st semester: Design, building, and integration of a position sensor of the kite. 2nd semester: Data acquisition and filter implementation (EKF, UKF) to localize the kite.	2024 – 2025 1 year
EPFL Rocket Team , Recovery System Engineer	Switzerland
Design and testing of the parachute of the Nordend rocket of the EPFL Rocket Team for the European EUROCC competition.	2022 – 2023 1 year

EPFL, Teaching Assistant

Switzerland

- Teaching assistant in Introduction to Computer Science, Analysis II, Fluid Mechanics, and Model Predictive Control.

2022–2023 / 2024–2025

Awards

Deep Learning Object Detection Competition

2025

Winner of the Deep Learning Object Detection competition 2025 in the Image Analysis and Pattern Recognition course at EPFL.

EPFL Space Race

2022

Winner EPFL Space Race 2022.

Certificates

Cambridge English Advanced Certificate

2020

English, C1 level.

Goeth Institut

2018

German B1.

Skills

Coding Languages: Strong experience in Python (PyTorch, Pandas, scikit-learn, ...). Worked on Matlab and knowledge in C/C++ (mostly for Arduino).

Languages

French: Mother tongue

English: C1

Dutch: Fluent

German: Scholar